
ALTm Tool v2.0

DNA-UTAH.ORG

Oligonucleotide Tm Prediction

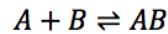
V2.0

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1 THERMODYNAMICS

T_m is the thermodynamic melting temperature of complementary DNA strands defined as the temperature where half of the strands form duplexes. The chemical reaction can be written as:



where A and B are complementary single strands, and AB is their duplex.

A) ΔG°

We start with two different definitions of Gibbs free energy ΔG° :

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta G^\circ = -RT\ln K$$

where: ΔH° is the change in enthalpy or heat of reaction,
 ΔS° is the change in entropy or disorder of the reaction,
T is temperature in Kelvin,
R is the gas constant, and
K is the equilibrium constant of the reaction
° indicates standard conditions

B) ELIMINATION OF ΔG°

Next, we eliminate ΔG° by combining the two equations above:

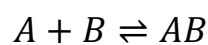
$$-RT\ln K = \Delta H^\circ - T\Delta S^\circ$$

$$\therefore T = \frac{\Delta H^\circ}{\Delta S^\circ - R\ln K}$$

Alternatively, the Arrhenius equation ($K = e^{-(\Delta H^\circ - T\Delta S^\circ)/RT}$) can be solved for T.

C) EXPRESSING THE EQUILIBRIUM CONSTANT (K) IN TERMS OF CONCENTRATION

For DNA, the two strands are either in the single strand (A, B) or double strand (AB) state.



Thus, the equilibrium constant K is:

$$K = \frac{[AB]}{[A][B]}$$

Substituting K into **B)** above:

$$T = \frac{\Delta H^\circ}{\Delta S^\circ - R \ln\left(\frac{[AB]}{[A][B]}\right)} = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln\left(\frac{[A][B]}{[AB]}\right)}$$

D) EVALUATING [A], [B], AND [AB] AT T_m

T_m is defined as the temperature at which 50% of the nucleic acid exists in the double strand state, where the '50' subscript indicates 50% of the maximum possible concentration.

The value of $\left(\frac{[A_{50}][B_{50}]}{[AB_{50}]}\right)$ depends on the initial concentrations of A and B.

#1 If [A]=[B]:

$$\left(\frac{[A_{50}][B_{50}]}{[AB_{50}]}\right) = \frac{[A]}{2}$$

[A]/2 is equivalent to ([A] + [B]) / 4, sometimes written as $\left(\frac{C_t}{4}\right)$ where C_t is the total strand concentration. A special case occurs with self-complimentary DNA strands, where A+A forms the duplex AA, and K = [AA]/[A][A]. At the T_m, half of the single strands remain ($[A_{50}] = \frac{C_t}{2}$), and the other half have duplexed ($[AA_{50}] = \frac{C_t}{4}$), so the concentration term becomes:

$$\frac{[A_{50}][A_{50}]}{[AA_{50}]} = \frac{C_t}{2} \times \frac{C_t}{2} \div \frac{C_t}{4} = C_t$$

#2 If [A] > [B]:

$$\left(\frac{[A_{50}][B_{50}]}{[AB_{50}]}\right) = \frac{\left([A] - \frac{[B]}{2}\right)\left(\frac{[B]}{2}\right)}{\left(\frac{[B]}{2}\right)} = [A] - \frac{[B]}{2}$$

E) T_m CALCULATIONS

There are four common scenarios for oligonucleotide T_m calculations, and each situation changes the appropriate calculation:

#1 Generic T_m calculations (unrelated to PCR)

When an oligonucleotide is not used as a primer or probe in PCR or any other amplification technique, the concentrations of both that oligonucleotide and its complementary sequence are important. In addition, monovalent cation concentrations (Na⁺, K⁺, Tris⁺) affect T_m, as well as the free Mg⁺⁺ concentration (calculated as the difference between the Mg⁺⁺ concentration and the total dNTP concentration).

#2 Primer T_m

At the beginning of PCR and other amplification methods, the primer concentration [A], is much greater than the template concentration [B]:

$$\begin{aligned}[A] - \frac{[B]}{2} &= [\textit{primer}] - \frac{[\textit{template}]}{2} \\ [\textit{primer}] &\gg [\textit{template}] \\ &= [\textit{primer}]\end{aligned}$$

That is, primer T_m only depends on the primer concentration, not the template concentration, as long as [primer]>>[template] which is true throughout most of PCR.

#3 Probe T_m for qPCR

During the exponential phase of PCR, probe concentration [A] is usually much greater than product concentration [B] and it is safe to assume that probe T_m depends only on the probe concentration:

$$\begin{aligned}[A] - [B]/2 &= [\textit{probe}] - [\textit{template}]/2 \\ [\textit{probe}] &\gg [\textit{template}] \\ &= [\textit{probe}]\end{aligned}$$

This is appropriate for probe design in qPCR.

#4 Probe T_m for melting analysis

After amplification, probe T_m depends on the probe concentration and the concentration of available target (PCR product).

a) Probe concentration. The probe concentration will decrease during PCR if a polymerase with 5'-exonuclease activity is used. This can be ignored during the exponential phase while the probe concentration is much greater than the product

concentration. However, for melting analysis at the end of PCR, significant degradation can occur and a default of 10% degradation is assumed.

b) Available product concentration. Probe T_m calculation at the end of PCR requires an estimate of the concentration of single-stranded product that is available for probe hybridization. This depends on whether the PCR is symmetric (same primer concentrations) or asymmetric (different primer concentrations).

i. *Symmetric PCR*. The concentration of single-stranded product available to hybridize to a probe depends on: 1) how much primer is converted to double-stranded product and 2) how much of that double-stranded product is available as single strands.

1) Primer conversion to product depends on the PCR efficiency and the number of PCR cycles.

2) The concentration of available single strands from double-stranded product depends on the cooling kinetics and annealing time.

Therefore, the [available product] = [primer] x (fraction converted to product) x (fraction of double-stranded product available as single strands). The default for both fractions is set to 0.5, but should be adjusted by the user based on the PCR conditions.

$$[\text{available product}] = [\text{primer}] * (f_{\text{prod}}) * (f_{\text{ds to ss}})$$

ii. *Asymmetric PCR*. In asymmetric PCR, the concentration of single-stranded product available to hybridize to the probe depends on: 1) the concentration of available single strands from double-stranded product, and 2) the concentration of single-stranded product produced by the excess primer.

1) The amount of double-stranded product after asymmetric PCR is equal to the concentration of the limiting primer. The fraction of available single strands from this double-stranded product depends on the cooling kinetics and annealing time.

2) The concentration of single strands produced by asymmetric PCR depends on the number of cycles of linear amplification after exponential PCR and will be a fraction of the maximum possible ([excess primer] – [limiting primer]).

Therefore, the [available product] = [limiting primer] x (fraction of double-stranded product available as single strands) + (fraction of the maximum possible concentration of single strands ([excess primer] – [limiting primer])). The default for both fractions is set to 0.5, but should be adjusted by the user based on the PCR conditions.

$$[\text{available product}] = [\text{primer}]_{\text{Lo}} * (f_{\text{ds to ss}}) + ([\text{primer}]_{\text{Hi}} - [\text{primer}]_{\text{Lo}}) * (f_{\text{max}})$$

Finally, for both symmetric and asymmetric PCR, if [available product] > [probe]:

$$[A] - \frac{[B]}{2} = [\text{available product}] - \frac{[\text{probe}]}{2}$$

If [probe] ≥ [available product]:

$$[A] - \frac{[B]}{2} = [probe] - \frac{[available\ product]}{2}$$

F) NEAREST NEIGHBOR MODEL

The nearest neighbor hypothesis of oligonucleotide thermodynamics assumes that ΔH° and ΔS° for the oligonucleotide can be approximated by the sum of all nearest neighbor pairs (NN).

$$\Delta H^\circ_{\text{oligo}} = \sum \Delta H^\circ = \Delta H^\circ_{\text{NN1}} + \Delta H^\circ_{\text{NN2}} + \dots + \Delta H^\circ_{\text{NNn}}$$

$$\Delta S^\circ_{\text{oligo}} = \sum \Delta S^\circ = \Delta S^\circ_{\text{NN1}} + \Delta S^\circ_{\text{NN2}} + \dots + \Delta S^\circ_{\text{NNn}}$$

From 16 permutations of two neighboring base pairs, 10 are unique:

```

→ → → → → → → → → →
AA AT TA CA GT CT GA CG GC GG
TT TA TA GT CA GA CT GC CG CC
← ← ← ← ← ← ← ← ← ←

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and 6 are redundant:

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→      → → → →      →
TT      TG AC AG TC      CC
AA      AC TG TC AG      GG
←      ← ← ← ←      ←

```

As a simple example:

$$\sum \Delta H^\circ \text{ of ATGCC} = \Delta H^\circ_{\text{AT}} + \Delta H^\circ_{\text{TG}} + \Delta H^\circ_{\text{GC}} + \Delta H^\circ_{\text{CC}}$$

For most accurate calculations, there are also initiation, terminal, and symmetry correction terms. In addition to the ten unique nearest neighbor pairs, values for mismatched pairs, terminal mismatches and dangling ends are available. In order to designate mismatches and dangling ends, we adopt the following nomenclature of SantaLucia (1998):

[Top strand (5' to 3')]/[Bottom strand (3' to 5')]. For example:

AC/TG is 5'-AC-3' paired with 3'-TG-5' (an AT match followed by a CG match)

AC/TT is 5'-AC-3' with 3'-TT-5' (an AT match followed by a CT mismatch)

AC/T- is 5'-AC-3' with 3'-T-5' (an AT match with a top strand C dangling end)

-C/TG is 5'-C-3' with 3'-TG-5' (a bottom strand T dangling end with a CG match)

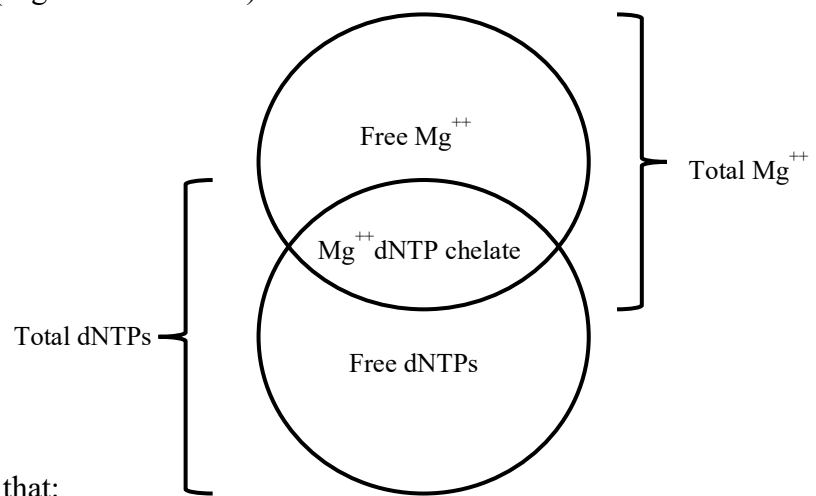
The ΔH° and ΔS° nearest neighbor parameters have been estimated by several groups and are tabulated in section 3.

G) SALT CORRECTION

Nearest neighbor thermodynamic values are often obtained in 1M NaCl, a condition not relevant to PCR. The concentration of salt affects the T_m and the major contributors are monovalent cations (in PCR these are usually Tris^+ and K^+) and the free Mg^{++} concentration. Monovalent cation concentrations are additive; the contributions of Na^+ , K^+ and Tris^+ (the ionized Tris at the pH used) are equal and can be summed. Free Mg^{++} is the concentration of Mg^{++} not bound to dNTPs. The [total dNTP] is equal to [dATP]+[dCTP]+[GTP]+[dTTP]. Typically, in PCR the [total dNTPs] is significantly less than [total Mg^{++}], and the [free Mg^{++}] can be estimated as [total Mg^{++}] – [total dNTPs]. However, as [total dNTPs] approaches the [total Mg^{++}], free Mg^{++} is more accurately calculated by considering the equilibrium constant as described by Owczarzy (2008) who estimated K_a as 30,000 M^{-1} :

(free Mg^{++}) + (free dNTPs) $\rightleftharpoons$ (Mg^{++} dNTP chelate) with:

$$K_a = \frac{[\text{Mg}^{++}\text{dNTP chelate}]}{[\text{free Mg}^{++}][\text{free dNTPs}]}$$



From the diagram, it is apparent that:

$$[\text{Mg}^{++}\text{dNTP chelate}] = [\text{Total Mg}^{++}] - [\text{free Mg}^{++}], \text{ and:}$$

$$[\text{free dNTPs}] = [\text{total dNTPs}] + [\text{free Mg}^{++}] - [\text{total Mg}^{++}], \text{ so that:}$$

$$30,000 = \frac{[\text{Total Mg}^{++}] - [\text{free Mg}^{++}]}{[\text{total dNTPs}] + [\text{free Mg}^{++}] - [\text{total Mg}^{++}]}$$

This equation can be solved for [free Mg^{++}] using the quadratic equation ($ax^2 + bx + c = 0$) with:

$$a = 30,000$$

$$b = 1 + 30,000 * ([\text{total dNTPs}] - [\text{total Mg}^{++}])$$

$$c = -[\text{total Mg}^{++}]$$

With $[\text{free Mg}^{++}] = \{-b + \text{SQRT}(b^2 - 4ac)\} / 2a$ (only one root is positive).

In summary, a good approximation for [free Mg^{++}] is [total Mg^{++}] – [total dNTPs] at most PCR concentrations. A more precise calculation of [free Mg^{++}] uses K_a and the quadratic equation and is used here.

Salt affects ΔS° , not ΔH° . There are 2 common additive corrections for $\sum \Delta S^\circ$:

#1 A simple additive term that modifies $\sum \Delta S^\circ$ is:

$$0.368 \times (N - 1) \times \ln([Mono^+] + 3.79\sqrt{[free Mg^{++}]})$$

where, N is the number of bases in the sequence. [Free Mg⁺⁺] can be calculated as [total Mg⁺⁺] - [total dNTPs], or more precisely using the K_a calculation detailed above.

This equation is derived from at least three sources. SantaLucia (1998) calculated the length dependence and log relationship to monovalent cations [Mono⁺] including Na⁺, K⁺ and Tris⁺. Mitsuhashi (1996) suggested the Na⁺ equivalence of Mg⁺⁺ scales as $\sqrt{[Mg^{++}]}$. Von Ahsen et al. (2001) determined experimental coefficients and introduced using free Mg⁺⁺ as the difference between total Mg⁺⁺ and the total dNTP concentrations. However, a more accurate calculation was empirically derived by Owczarzy and is used here.

#2 Owczarzy et al. (2008) published a salt correction that depends additionally on the fractional GC content (f_{GC}) and $\sum\Delta H^\circ$ (multiplied by 1,000 to convert from kcal/mol to cal/mol). There are 3 cases depending on the value of $\frac{\sqrt{[free Mg^{++}]}}{[Mono^+]}$:

a) For $\frac{\sqrt{[free Mg^{++}]}}{[Mono^+]} < 0.22$, the term depends on f_{GC} and [Mono⁺]:

$$1000\sum\Delta H^\circ [(4.29 f_{GC} - 3.95) \times 10^{-5} \times \ln[Mono^+] + 9.40 \times 10^{-6} \times (\ln[Mono^+])^2]$$

b) For $0.22 \leq \frac{\sqrt{[free Mg^{++}]}}{[Mono^+]} < 6.0$, including almost all PCR conditions, the term depends on F_{GC}, N, [Mono⁺] and [free Mg⁺⁺]:

$$1000\sum\Delta H^\circ \{a + b \ln[free Mg^{++}] + f_{GC}(c + d \ln[free Mg^{++}]) + \frac{1}{2(N-1)} [e + f \ln[free Mg^{++}] + g(\ln[free Mg^{++}])^2]\}$$

with coefficients:

$$a = 3.92 \times 10^{-5} (0.843 - 0.352\sqrt{[Mono^+]} \times \ln[Mono^+])$$

$$b = -9.11 \times 10^{-6}$$

$$c = 6.26 \times 10^{-5}$$

$$d = 1.42 \times 10^{-5} (1.279 - 4.03 \times 10^{-3} \ln[Mono^+] - 8.03 \times 10^{-3} (\ln[Mono^+])^2)$$

$$e = -4.82 \times 10^{-4}$$

$$f = 5.25 \times 10^{-4}$$

$$g = 8.31 \times 10^{-5} (0.486 - 0.258 \ln[Mono^+] + 5.25 \times 10^{-3} (\ln[Mono^+])^3)$$

c) For $\frac{\sqrt{[free Mg^{++}]}}{[Mono^+]} \geq 6.0$, the term depends on f_{GC} , N and $[free Mg^{++}]$:

$$1000\sum\Delta H^\circ\{a + b \ln[free Mg^{++}] + f_{GC}(c + d \ln[free Mg^{++}]) + \frac{1}{2(N-1)}[e + f \ln[free Mg^{++}] + g(\ln[free Mg^{++}])^2]\}$$

with coefficients:

$$a = 3.92 \times 10^{-5}$$

$$b = -9.11 \times 10^{-6}$$

$$c = 6.26 \times 10^{-5}$$

$$d = 1.42 \times 10^{-5}$$

$$e = -4.82 \times 10^{-4}$$

$$f = 5.25 \times 10^{-4}$$

$$g = 8.31 \times 10^{-5}$$

2 FULL T_m EQUATION

- T_m in °C
- ΣΔH°, in kcal/mol, is the sum of all nearest neighbor ΔH terms for the oligo
- ΣΔS°, in cal/(mol*K) or cal/K/mol, is the sum of all nearest neighbor ΔS terms for the oligo
- Salt Correction depends on [free Mg⁺⁺] and [total monovalent cations] (see section 1G))
- 1.9872 is the gas constant R in cal/(mol*K)
- 1000 converts kcal/mol to cal/mol for ΣΔH°
- -273.15 converts K to °C
- [A] and [B] are the initial molar concentrations of the two hybridizing strands with [A] ≥ [B]

$$T_m = \frac{\Sigma\Delta H^\circ \times 1000}{\Sigma\Delta S^\circ + (\text{Salt Correction}) + 1.9872 \times \ln([A] - \frac{[B]}{2})} - 273.15$$

For primer T_m:

$$\begin{aligned}[A] &= [\text{primer}] \\ [B] &= 0\end{aligned}$$

For probe T_m during the exponential phase:

$$\begin{aligned}[A] &= [\text{probe}] \\ [B] &= 0\end{aligned}$$

For probe T_m at the end of PCR (melting analysis):

For symmetric PCR:

$$\begin{aligned}\text{With } [\text{probe}] < [\text{available product}]: \\ [A] &= [\text{primer}] * (f_{\text{prod}}) * (f_{\text{ds to ss}}) \\ [B] &= [\text{probe}]\end{aligned}$$

With [probe] ≥ [available product]:

$$\begin{aligned}[A] &= [\text{probe}] \\ [B] &= [\text{primer}] * (f_{\text{prod}}) * (f_{\text{ds to ss}})\end{aligned}$$

For asymmetric PCR:

With [probe] < [available product]:

$$\begin{aligned}[A] &= [\text{primer}]_{\text{Lo}} * (f_{\text{ds to ss}}) + ([\text{primer}]_{\text{Hi}} - [\text{primer}]_{\text{Lo}}) * (f_{\text{max}}) \\ [B] &= [\text{probe}]\end{aligned}$$

With [probe] ≥ [available product]:

$$\begin{aligned}[A] &= [\text{probe}] \\ [B] &= [\text{primer}]_{\text{Lo}} * (f_{\text{ds to ss}}) + ([\text{primer}]_{\text{Hi}} - [\text{primer}]_{\text{Lo}}) * (f_{\text{max}})\end{aligned}$$

See section E#4b for variable definitions.

3 NEAREST NEIGHBOR (NN) PARAMETERS

A) Ten complementary nearest neighbor pairs

SantaLucia (1998):

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TT or TT/AA	ATpAT or TApTA	-7.9	-22.2
AT/TA	ATpTA	-7.2	-20.4
TA/AT	TApAT	-7.2	-21.3
CA/GT or TG/AC	CGpAT or TApGC	-8.5	-22.7
GT/CA or AC/TG	GCpTA or ATpCG	-8.4	-22.4
CT/GA or AG/TC	CGpTA or ATpGC	-7.8	-21.0
GA/CT or TC/AG	GCpAT or TApCG	-8.2	-22.2
CG/GC	CGpGC	-10.6	-27.2
GC/CG	GCpCG	-9.8	-24.4
GG/CC or CC/GG		-8.0	-19.9
Init & terminal GC		0.1	-2.8
Init & terminal AT		2.3	4.1
Symmetry correction		0	-1.4

Each sequence has both 5' and 3' termini. If both termini are GC, the "Init & terminal GC" term is added twice, etc. The symmetry correction is only applied in the rare case of a self-complementary oligonucleotide.

SantaLucia and Hicks (2004):

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TT or TT/AA	ATpAT or TApTA	-7.6	-21.3
AT/TA	ATpTA	-7.2	-20.4
TA/AT	TApAT	-7.2	-21.3
CA/GT or TG/AC	CGpAT or TApGC	-8.5	-22.7
GT/CA or AC/TG	GCpTA or ATpCG	-8.4	-22.4
CT/GA or AG/TC	CGpTA or ATpGC	-7.8	-21.0
GA/CT or TC/AG	GCpAT or TApCG	-8.2	-22.2
CG/GC	CGpGC	-10.6	-27.2
GC/CG	GCpCG	-9.8	-24.4
GG/CC or CC/GG	ATpAT or TApTA	-8.0	-19.9
Initiation		0.2	-5.7
Terminal AT penalty		2.2	6.9
Symmetry correction		0.0	-1.4

The terminal AT penalty is applied to each end (if both ends are AT, the penalty is added twice). **X.X** values are different from SantaLucia (1998).

NNDB: https://rna.urmc.rochester.edu/NNDB/dna/dna_watson_crick_helices.html

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TT or TT/AA	ATpAT or TApTA	-1.0	-7.2	-20.0
AT/TA	ATpTA	-0.9	-5.3	-14.2
TA/AT	TApAT	-0.6	-6.7	-19.7
CA/GT or TG/AC	CGpAT or TApGC	-1.5	-9.9	-27.1
GT/CA or AC/TG	GCpTA or ATpCG	-1.4	-7.8	-20.6
CT/GA or AG/TC	CGpTA or ATpGC	-1.3	-5.8	-14.5
GA/CT or TC/AG	GCpAT or TApCG	-1.3	-8.5	-23.2
CG/GC	CGpGC	-2.2	-9.8	-24.5
GC/CG	GCpCG	-0.6	-2.4	-5.8
GG/CC or CC/GG	ATpAT or TApTA	-0.3	-7.5	-23.2
Initiation		1.0	-7.2	-26.4
Terminal AT penalty		0.0	3.2	10.3
Symmetry correction		4.5	0.0	-14.5

B) Mismatch Nearest Neighbor Pairs (not terminal or penultimate)

1) G/T mismatches

Single mismatches: (Allawi and SantaLucia, 1997)

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AG/TT or TT/GA	ATpGT or TGpTA	1.0	0.9
AT/TG or GT/TA	ATpTG or GTpTA	-2.5	-8.3
CG/GT or TG/GC	CGpGT or TGpGC	-4.1	-11.7
CT/GG or GG/TC	CGpTG or GTpGC	-2.8	-8.0
GG/CT or TC/GG	GCpGT or TGpCG	3.3	10.4
GT/CG or GC/TG	GCpTG or GTpCG	-4.4	-12.3
TG/AT or TA/GT	TApGT or TGpAT	-0.1	-1.7
TT/AG or GA/TT	TApTG or GTpAT	-1.3	-5.3

Single mismatches:

https://rna.urmc.rochester.edu/NNDB/dna/dna_watson_crick_helices.html

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AG/TT or TT/GA	ATpGT or TGpTA	0.7	3.7	9.7
AT/TG or GT/TA	ATpTG or GTpTA	0.1	-1.7	-5.8
CG/GT or TG/GC	CGpGT or TGpGC	-0.5	-4.3	-12.3
CT/GG or GG/TC	CGpTG or GTpGC	-0.3	-2.4	-6.8
GG/CT or TC/GG	GCpGT or TGpCG	0.1	4.3	13.5
GT/CG or GC/TG	GCpTG or GTpCG	-0.6	-2.4	-5.8
TG/AT or TA/GT	TApGT or TGpAT	0.4	-0.9	-4.2
TT/AG or GA/TT	TApTG or GTpAT	0.3	-0.9	-3.9

Double mismatches: (Allawi and SantaLucia, 1997)

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
GG/TT or TT/GG	GT/GT or TGpTG	5.8	16.3
GT/TG	GTpTG	4.1	9.5
TG/GT	TGpGT	-1.4	-6.2

Double mismatches:

https://rna.urmc.rochester.edu/NNDB/dna/dna_watson_crick_helices.html

NN Sequence (5'→3')/(3'→5')	Weber Notation (1 st bp)p(2 nd bp)	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
GG/TT or TT/GG	GT/GT or TGpTG	0.7	8.0	23.5
GT/TG	GTpTG	1.2	5.0	12.3
TG/GT	TGpGT	0.5	-6.7	-23.2

2) G/A mismatches (Allawi and SantaLucia, 1998)

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TG or GT/AA	-0.6	-2.3
AG/TA or AT/GA	-0.7	-2.3
CA/GG or GG/AC	-0.7	-2.3
CG/GA or AG/GC	-4.0	-13.2
GA/CG or GC/AG	-0.6	-1.0
GG/CA or AC/GG	0.5	3.2
TA/AG or GA/AT	0.7	0.7
TG/AA or AA/GT	3.0	7.4

3) A/C mismatches, pH 7.0 (Allawi and SantaLucia, 1998)

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TC or CT/AA	2.3	4.6
AC/TA or AT/CA	5.3	14.6
CA/GC or CG/AC	1.9	3.7
CC/GA or AG/CC	0.6	-0.6
GA/CC or CC/AG	5.2	14.2
GC/CA or AC/CG	-0.7	-3.8
TA/AC or CA/AT	3.4	8.0
TC/AA or AA/CT	7.6	20.2

4) C/T mismatches (Allawi and SantaLucia, 1998)

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AC/TT or TT/CA	0.7	0.2
AT/TC or CT/TA	-1.2	-6.2
CC/GT or TG/CC	-0.8	-4.5
CT/GC or CG/TC	-1.5	-6.1
GC/CT or TC/CG	2.3	5.4
GT/CC or CC/TG	5.2	13.5
TC/AT or TA/CT	1.2	0.7
TT/AC or CA/TT	1.0	0.7

5) A/A, C/C, G/G, and T/T mismatches (Allawi and SantaLucia, 1999)

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TA or AT/AA	1.2	1.7
CA/GA or AG/AC	-0.9	-4.2
GA/CA or AC/AG	-2.9	-9.8
TA/AA or AA/AT	4.7	12.9
AC/TC or CT/CA	0.0	-4.4

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
CC/GC or CG/CC	-1.5	-7.2
GC/CC or CC/CG	3.6	8.9
TC/AC or CA/CT	6.1	16.4
AG/TG or GT/GA	-3.1	-9.5
CG/GG or GG/GC	-4.9	-15.3
GG/CG or GC/GG	-6.0	-15.8
TG/AG or GA/GT	1.6	3.6
AT/TT or TT/TA	-2.7	-10.8
CT/GT or TG/TC	-5.0	-15.8
GT/CT or TC/TG	-2.2	-8.4
TT/AT or TA/TT	0.2	-1.5

C) Dangling end parameters (Bommarito, 2000)

1) 5'-dangling ends:

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/-T or T/-AA	0.2	2.3
AC/-G or G/-CA	-6.3	-17.1
AG/-C or C/-GA	-3.7	-10.0
AT/-A or A/-TA	-2.9	-7.6
CA/-T or T/-AC	0.6	3.3
CC/-G or G/-CC	-4.4	-12.6
CG/-C or C/-GC	-4.0	-11.9
CT/-A or A/-TC	-4.1	-13.0
GA/-T or T/-AG	-1.1	-1.6
GC/-G or G/-CG	-5.1	-14.0
GG/-C or C/-GG	-3.9	-10.9
GT/-A or A/-TG	-4.2	-15.0
TA/-T or T/-AT	-6.9	-20.0
TC/-G or G/-CT	-4.0	-10.9
TG/-C or C/-GT	-4.9	-13.8
TT/-A or A/-TT	-0.2	-0.5

2) 3'-dangling ends:

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
-A/TA or AT/A-	-0.7	-0.8
-C/GA or AG/C-	-2.1	-3.9
-G/CA or AC/G-	-5.9	-16.5
-T/AA or AA/T-	-0.5	-1.1

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
-A/TC or CT/A-	4.4	7.3
-C/GC or CG/C-	-0.2	-0.1
-G/CC or CC/G-	-2.6	-7.4
-T/AC or CA/T-	4.7	14.2
-A/TG or GT/A-	-1.6	-3.6
-C/GG or GG/C-	-3.9	-11.2
-G/CG or GC/G-	-3.2	-10.4
-T/AG or GA/T-	-4.1	-13.1
-A/TT or TT/A-	2.9	10.4
-C/GT or TG/C-	-4.4	-13.1
-G/CT or TC/G-	-5.2	-15.0
-T/AT or TA/T-	-3.8	-12.6

D) Deoxyinosine (Watkins and SantaLucia, 2005)

1) I/C pairs

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AI/TC or CT/IA	-8.9	-25.5
TI/AC or CA/IT	-5.9	-17.4
AC/TI or IT/CA	-8.8	-25.4
TC/AI or IA/CT	-4.9	-13.9
CI/GC or CG/IC	-5.4	-13.7
GI/CC or CC/IG	-6.8	-19.1
CC/GI or IG/CC	-8.3	-23.8
GC/CI or IC/CG	-5.0	-12.6

2) I/A pairs

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AI/TA or AT/IA	-8.3	-25.0
TI/AA or AA/IT	-3.4	-11.2
AA/TI or IT/AA	-0.7	-2.6
TA/AI or IA/AT	-1.3	-4.6
CI/GA or AG/IC	2.6	8.9
GI/CA or AC/IG	-7.8	-21.1
CA/GI or IG/AC	-7.0	-20.0
GA/CI or IC/AG	-7.6	-20.2

3) I/T pairs

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AI/TT or TT/IA	0.49	-0.7
TI/AT or TA/IT	-6.5	-22.0
AT/TI or IT/TA	-5.6	-18.7
TT/AI or IA/TT	-0.8	-4.3
CI/GT or TG/IC	-1.0	-2.4
GI/CT or TC/IG	-3.5	-10.6
CT/GI or IG/TC	0.1	-1.0
GT/CI or IC/TG	-4.3	-12.1

4) I/G pairs

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AI/TG or GT/IA	-4.9	-15.8
TI/AG or GA/IT	-1.9	-8.5
AG/TI or IT/GA	0.1	-1.8
TG/AI or IA/GT	1.0	1.0
CI/GG or GG/IC	7.1	21.3
GI/CG or GC/IG	-1.1	-3.2
CG/GI or IG/GC	5.8	16.9
GG/CI or IC/GG	-7.6	-22.0

5) I/I pairs

NN Sequence (5'→3')/(3'→5')	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AI/II or II/IA	-3.3	-11.9
TI/II or II/IT	0.1	-2.3
CI/GI or IG/IC	1.3	3.0
GI/CI or IC/IG	-0.5	-1.3

E) Hairpin Loops – Two databases are available:

1. https://rna.urmc.rochester.edu/NNDB/dna/dna_hairpin_loops.html

Initiation of loop size N

Loop Size	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
3	3.4	-0.6	-12.9
4	3.4	-2.3	-18.4
5	3.5	-12.1	-50.3
6	4.2	-16.7	-67.4
7	4.2	-13.6	-57.4
8	4.2	-14.1	-59.0
9	4.3	-14.1	-59.3
10	4.4	-14.1	-59.6
11	4.5	-14.1	-60.0
12	4.6	-14.1	-60.3
13	4.7	-14.1	-60.6
14	4.8	-14.1	-60.9
15	5.0	-14.1	-61.6
16	5.1	-14.1	-61.9
17	5.2	-14.1	-62.2
18	5.3	-14.1	-62.6
19	5.4	-14.1	-62.9
20	5.5	-14.1	-63.2
21	5.6	-14.1	-63.5
22	5.7	-14.1	-63.8
23	5.8	-14.1	-64.2
24	5.9	-14.1	-64.5
25	6.0	-14.1	-64.8
26	6.1	-14.1	-65.1
27	6.2	-14.1	-65.5
28	6.3	-14.1	-65.8
29	6.4	-14.1	-66.1
30	6.5	-14.1	-66.4

For loop sizes (N) greater than 30 bases:

$$\Delta H^{\circ} = -14.1$$

$$\Delta G^{\circ}_{37} = 6.5 + 0.1(N-30)$$

$$\Delta S^{\circ} \text{ (cal/mol}^{\circ}\text{K)} = - [14.1 + 6.5 + 0.1(N-30)]1000/310.15$$

2. Santa Lucia and Hicks (2004)

Assumes $\Delta H^\circ = 0$

We corrected original errors in SantaLucia and Hicks (2004) as follows:

Table 4, footnote a: $\Delta S^\circ = \Delta G^\circ (1000)/310.15$

Equation 7: $\Delta G^\circ(\text{loop-n}) = \Delta G^\circ(\text{loop-x}) + (2.44)R(310.15)\ln(n/x)/1000$

The table below reflects these corrections:

Loop sizes up to 30 (if not shown, interpolate)

Loop Size	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
3	3.5	0	-11.3
4	3.5	0	-11.3
5	3.3	0	-10.6
6	4.0	0	-12.9
7	4.2	0	-13.5
8	4.3	0	-13.9
9	4.5	0	-14.5
10	4.6	0	-14.8
12	5.0	0	-16.1
14	5.1	0	-16.4
16	5.3	0	-17.1
18	5.5	0	-17.7
20	5.7	0	-18.4
25	6.1	0	-19.7
30	6.3	0	-20.3

For loop sizes (N) greater than 30 bases:

$$\begin{aligned}
 \Delta G^\circ \text{ (kcal/mol)} &= 6.3 + [(2.44)R(310.15)/1000]\ln(n/30) \\
 &= 6.3 + (2.44)(1.9872)(0.31015)\ln(n/30) \\
 &= 6.3 + 1.50\ln(n/30) \\
 \Delta S^\circ \text{ (cal/mol*K)} &= - [6.3 + 1.50\ln(n/30)](1000)/310.15
 \end{aligned}$$

F) Hairpin Loops - Snapback primer ΔH° and ΔS° calculations:

$$\Delta H^\circ_{(\text{snapback})} =$$

$\Delta H^\circ(\text{stem})$ – standard NN calculations

$-\Delta H^\circ(\text{initiation stem}) = 0.2 \text{ kcal/mol}$

$+\Delta H^\circ(\text{terminal mismatch of tail designed to prevent extension of minor product})$

$+\Delta H^\circ(\text{terminal mismatch of internal loop})$

$+\Delta H^\circ(\text{initiation of loop if size N})$

$$\Delta S^\circ_{(\text{snapback})} =$$

$\Delta S^\circ(\text{stem})$ – standard NN calculations

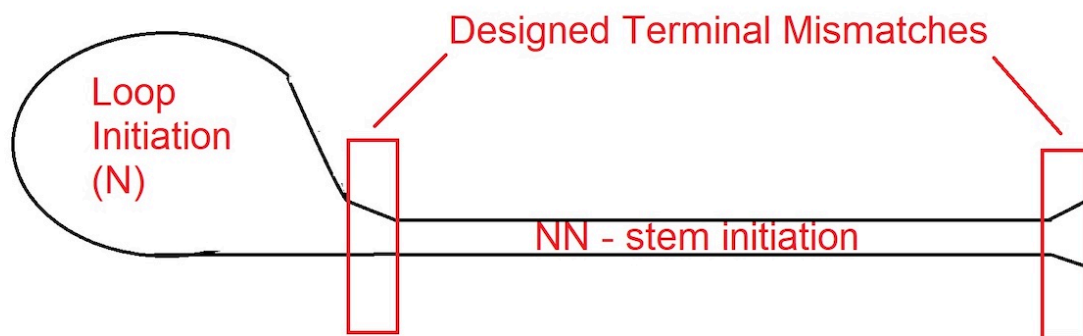
$-\Delta S^\circ(\text{initiation stem}) = -5.7 \text{ cal/mol}$

$+\Delta S^\circ(\text{terminal mismatch of tail designed to prevent extension of minor product})$

$+\Delta S^\circ(\text{terminal mismatch of internal loop})$

$+\Delta S^\circ(\text{initiation of loop of size N})$

$$T_{m(\text{snapback})} = \frac{\sum \Delta H^\circ(\text{snapback}) \times 1000}{\sum \Delta S^\circ(\text{snapback}) + (\text{Salt Correction})} - 273.15$$



G) Terminal Mismatches:

(https://rna.urmc.rochester.edu/NNDB/dna/dna_terminal_mismatches.html)

A/T neighbor match:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
AA/TA or AT/AA	-0.7	4.0	15.2
AA/TC or CT/AA	-0.3	5.2	17.7
AA/TG or GT/AA	-0.5	4.7	16.8
AC/TA or AT/CA	-0.6	4.0	14.8
AC/TC or CT/CA	-0.2	5.2	17.4
AC/TT or TT/CA	-0.3	4.3	14.8
AG/TA or AT/GA	-0.6	4.0	14.8
AG/TG or GT/GA	-0.4	4.7	16.4
AG/TT or TT/GA	-0.5	4.3	15.5
AT/TC or CT/TA	-0.3	5.2	17.7
AT/TG or GT/TA	-0.5	4.7	16.8
AT/TT or TT/TA	-0.4	4.3	15.2

T/A neighbor match:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
TA/AA or AA/AT	-0.6	2.3	9.4
TA/AC or CA/AT	-0.4	1.7	6.8
TA/AG or GA/AT	-0.5	0.7	3.9
TC/AA or AA/CT	-0.5	2.3	9.0
TC/AC or CA/CT	-0.2	1.7	6.1
TC/AT or TA/CT	-0.5	-5.8	-17.1
TG/AA or AA/GT	-0.6	2.3	9.4
TG/AG or GA/GT	-0.4	0.7	3.5
TG/AT or TA/GT	-0.5	-5.8	-17.1
TT/AC or CA/TT	-0.3	1.7	6.4
TT/AG or GA/TT	-0.6	0.7	4.2
TT/AT or TA/TT	-0.3	-5.8	-17.7

C/G neighbor match:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
CA/GA or AG/AC	-1.0	-4.6	-11.6
CA/GC or CG/AC	-0.8	-4.2	-11.0
CA/GG or GG/AC	-0.9	-4.7	-12.3
CC/GA or AG/CC	-0.8	-4.6	-12.3
CC/GC or CG/CC	-0.5	-4.2	-11.9
CC/GT or TG/CC	-0.7	-5.0	-13.9

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
CG/GA or AG/GC	-1.0	-4.6	-11.6
CG/GG or GG/GC	-0.9	-4.7	-12.3
CG/GT or TG/GC	-1.0	-5.0	-12.9
CT/GC or CG/TC	-0.6	-4.2	-11.6
CT/GG or GG/TC	-0.9	-4.7	-12.3
CT/GT or TG/TC	-0.9	-5.0	-13.2

G/C neighbor match:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
GA/CA or AC/AG	-1.0	-6.2	-16.8
GA/CC or CC/AG	-0.7	-3.3	-8.4
GA/CG or GC/AG	-0.8	-5.1	-13.9
GC/CA or AC/CG	-1.0	-6.2	-16.8
GC/CC or CC/CG	-0.6	-3.3	-8.7
GC/CT or TC/CG	-0.7	-0.9	-0.6
GG/CA or AC/GG	-1.0	-6.2	-16.8
GG/CG or GC/GG	-1.0	-5.1	-13.2
GG/CT or TC/GG	-0.8	-0.9	-0.3
GT/CC or CC/TG	-0.6	-3.3	-8.7
GT/CG or GC/TG	-0.9	-5.1	-13.5
GT/CT or TC/TG	-0.9	-0.9	0.0

H) Penultimate G/T Mismatches with Terminal Mismatch

(https://rna.urmc.rochester.edu/NNDB/dna/dna_terminal_mismatches.html)

5'-penultimate G/T with terminal mismatch:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
GA/TA or AT/AG	-0.5	-3.0	-8.1
GA/TC or CT/AG	-0.2	-0.1	0.3
GA/TG or GT/AG	-0.5	-1.9	-4.5
GC/TA or AT/CG	-0.2	-3.0	-9.0
GC/TC or CT/CG	-0.2	-0.1	0.3
GC/TT or TT/CG	-0.2	2.3	8.1
GG/TA or AT/GG	-0.5	-3.0	-8.1
GG/TG or GT/GG	-0.5	-1.9	-4.5
GG/TT or TT/GG	-0.5	2.3	9.0
GT/TC or CT/TG	-0.2	-0.1	0.3
GT/TG	-0.5	-1.9	-4.5
GT/TT or TT/TG	-0.2	2.3	8.1

5'-penultimate T/G with terminal mismatch:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
TA/GA or AG/AT	-0.5	2.3	9.0
TA/GC or CG/AT	-0.2	1.7	6.1
TA/GG or GG/AT	-0.5	0.7	3.9
TC/GA or AG/CT	-0.2	2.3	8.1
TC/GC or CG/CT	-0.2	1.7	6.1
TC/GT or TG/CT	-0.2	-5.8	-18.1
TG/GA or AG/GT	-0.5	2.3	9.0
TG/GG or GG/GT	-0.5	0.7	3.9
TG/GT	-0.5	-5.8	-17.1
TT/GC or CG/TT	-0.2	1.7	6.1
TT/GG or GG/TT	-0.5	0.7	3.9
TT/GT or TG/TT	-0.2	-5.8	-18.1

Note identity of tetrads in blue highlight with reported different ΔH° values. For both sequences, the average $\Delta H^{\circ} = 1.5$ with calculated average $\Delta S^{\circ} = 6.5$ is used.

Penultimate T/G with terminal match:

NN Sequence (5'→3')/(3'→5')	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/K/mol)
GA/TT or TT/AG	-0.6	2.3	9.4
GC/TG or GT/CG	-0.9	-1.9	-3.2
GG/TC or CT/GG	-0.9	-0.1	2.6
GT/TA or AT/TG	-0.5	-3	-8.1
TA/GT or TG/AT	-0.2	-5.8	-18.1
TC/GG or GG/CT	-0.8	0.7	4.8
TG/GC or CG/GT	-1	1.7	8.7
TT/GA or AG/TT	-0.5	2.3	9.0

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